Dilución $\quad Mc \cdot Vc = Md \cdot Vd$.

$\downarrow$

%.m/v.

7.34 SL 0,750 M KOH $\longrightarrow$ 0,150M KOH

V? $\qquad$ 1000 cm³

$$Mc \cdot Vc = Md \cdot Vd \Rightarrow Vc = \frac{\overset{0,75}{Md} \cdot \overset{1000cm^3}{Vd}}{\underset{0,15}{Mc}}$$

7.39 0,5L Vd $\longrightarrow$ 5cm³ = Vc $\quad$ Cc $\quad \rho = 1,84 \frac{g}{cm^3}$ $\quad$ 98 %.m/m.

Cd [%.m/v]

$$Vd\ Md = Mc\ Vc.$$

M = concentración.

Vd: L; Md: %.m/v; Mc: %.m/m; Vc: L

%.m/m $\xrightarrow{\rho}$ %.m/v.

Vd = 500 cm³ $\qquad$ Cc = Mc = 98%.m/m = 98g sto → 100g SN

Vc = 5 cm³ $\qquad$ m/v = x g sto → 100ml SN

Cc = 180,3 %.m/v $\qquad$ $\rho = \frac{m}{V} \Rightarrow V = \frac{m}{\rho} \Rightarrow V = \frac{100}{1,84} = 54,35ml$

$\Downarrow$ $\qquad$ 98g sto → 54,3ml SN

$$Cd = \frac{Cc \cdot Vc}{Vd}.$$

180,3g = X ← 100 ml SN

$$Cd = \frac{180,3 \cdot 5}{500} \Rightarrow Cd = 1,8\ \%.m/v.$$

Escaneado con CamScanner

8.14

$$8\,Hg(l) + S_8(s) \longrightarrow 8\,HgS(s)$$

2 ml masa?

$\rho = 13{,}6\ g/mL$

	Hg	S	HgS
g/mol PM	200,6	32	232,6
mol ne	8	1	8
masa me	1604,8	32	1860,8

V_0 2 ml

ρ_d 13,6 g/ml

m_d 27,2 g. ?

1604,8 g Hg → 8 · 32 S8.

27,2 g Hg → ~~0,54~~ 4,34 g.

8.16

$$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$$

masa? 0,8 M V?

2,5 L
20 °C
1 atm

$PV = nRT$

$n = \frac{PV}{RT} = \frac{1\,atm \cdot 2{,}5\,L}{0{,}082\,\frac{atm\,L}{K\,mol} \cdot 293\,K}$

$n = 0{,}104\ mol$

	Zn	HCl	$ZnCl_2$	H_2
PM	65,4	36,5	136,4	2
ne	1	2	1	1
me	65,4	73	136,4	2
n_T	0,104	0,208		0,104
m_T	6,8 g			0,208

2 g H_2 → 65,4 g Zn

0,208 g H_2 → x = 6,8 g Zn

0,8 M = 0,8 mol → 1000 ml SN.

0,208 mol → x = 260 ml SN

Escaneado con CamScanner

8.21

$$6\,NaNO_2 + 3\,H_2SO_4 \rightarrow 4\,NO + 2\,HNO_3 + 2\,H_2O + 3\,Na_2SO_4$$

100g	2M	10L	?
P=90%	V?	20°C	PV=nRT
		P?	

$$P = \frac{nRT}{V}$$

$$P = \frac{0{,}87\,mol \; 0{,}082\,atm\,L \cdot 293\,K}{10\,L \quad K\,mol}$$

$$P = 2{,}09\,atm$$

	$NaNO_2$	H_2SO_4	NO
PM	69,1 g/mol	98 g/mol	30 g/mol
ne	6 mol	3 mol	4 mol
me	414 g	294 g	120 g.
m_T	90g		
n_T	1,304 mol	0,652 mol	0,87 mol ✓

2M = 2 mol sto → 1000 ml SN
0,652 mol → X = 326 ml SN.

8.25

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2\,HCl$$

42,5g	exceso	40g.
	0,9M	

	$BaCl_2$	H_2SO_4	$BaSO_4$
PM	208,3 g/mol	98 g/mol	233,3 g/mol
ne	1 mol	1 mol	1 mol
me	208,3 g	98 g	233,3 g.
m_T	42,5g	16,8g	40g.
n_T	35,71	0,171 mol	

0,9 mol → 1000 ml SN
0,171 mol → X = 190 cm³

233,3g $BaSO_4$ → 208,3g $BaCl_2$
40g $BaSO_4$ → X = 35,71 g $BaCl_2$

42,5 g → 100%
35,71 g → 84%

35,71g $BaCl_2$ → X = 16,8.
208,3g $BaCl_2$ → 98g

~~35,71g → X =~~
~~98g ← 208,3g.~~

8.30

$$Zn_3Sb_2 + 6\,HCl \rightarrow 2\,SbH_3 + 3\,ZnCl_2$$

17kg	exceso	mona?
	R=14%	

	Zn_3Sb_2	SbH_3
PM	439,7 g/mol	124,75 g/mol
ne	1 mol	2 mol
me	439,7g	249,5 g
m_T	17000g	

439,7 g → 249,5g
17000 g → X = 9646,35 g

100% → 9646,35g
14% → X = 1350,5g ✓

Escaneado con CamScanner

8.32 $(NH_4)_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3NH_3 + 3H_2O$.

RL { 430 g, P = 80 %, 344 g, 2,3 mol } | 5 L, 2 M, 10 mol | R = 75 % | V?, T = 20 ºC, P = 1,50 atm.

	$(NH_4)_3PO_4$	NaOH	NH_3
PM	149 g/mol	40 g/mol	17 g/mol
ne	1	3	3
me	149 g	120 g	51 g.
m_T	344 g		
n_T	2,3 mol		

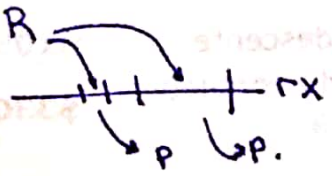
2M: 2 mol → 1000 ml SN
10 mol ← 5000 ml SN

1 mol $(NH_4)_3PO_4$ + 3 mol NaOH.
3,33 mol $(NH_4)_3PO_4$ + 10 mol NaOH
2,3 mol $(NH_4)_3PO_4$ + 6,9 mol NaOH

6,9 mol (100 %)
5,175 mol (75 %)

$$PV = nRT \Rightarrow V = \frac{nRT}{P} \Rightarrow V = \frac{5,175\ mol \cdot 0,082\ \frac{atm\ L}{mol\ K} \cdot 293\ K}{1,5\ atm} \quad V = 82,88\ L.$$

CINÉTICA QUÍMICA



$A \rightarrow B$.

$$r = \frac{\Delta[A]}{\Delta t} \qquad r_A = -r_B$$

$$r = \frac{\Delta[B]}{\Delta t} \qquad -r_A = r_B.$$

$$aA + bB \rightarrow cC + dD$$

$$r = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\cdot\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}.$$

$$r = k[A]^{\alpha}[B]^{\beta}$$

$r = kC^0$ $\quad k = \frac{M}{s}$. $\qquad r = kC^2$. $\quad k = \frac{L}{mol\,s}$

$r = kC^1$ $\quad k = \frac{1}{s}$ $\qquad r = kC^3$ $\quad k = \frac{L^2}{mol^2 s}$.

9.1 a) $H_2 + I_2 \rightarrow 2HI$

$$r = -\frac{1}{1}\frac{\Delta[H_2]}{\Delta t} = -\frac{1}{1}\cdot\frac{\Delta[I_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[HI]}{\Delta t}.$$

Escaneado con CamScanner

9.5 $2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$.

$r = k[NO]^{\alpha=2}[H_2]^{\beta=1}$

(a) $r = k[NO]^2[H_2]$

(b) $r = 15\ M/s$
$[NO] = 0{,}05M$
$[H_2] = 0{,}1M$

$r = k[NO]^2[H_2]$.

$\frac{r}{[NO]^2[H_2]} = k$.

9.7 $\underbrace{(CH_3)_3CBr}_{A} + \underbrace{OH^-}_{B} \rightarrow (CH_3)_3COH + Br^-$

$r = k[A]^\alpha[B]^\beta$. (a) determinar α, β, k.

Exp1
Exp2

$r_1 = k[A]_1^\alpha[B]_1^\beta$
$r_2 = k[A]_2^\alpha[B]_2^\beta$

$\frac{r_1}{r_2} = \frac{\cancel{k}[A]_1^\alpha \cancel{[B]_1^\beta}}{\cancel{k}[A]_2^\alpha \cancel{[B]_2^\beta}}$ $[B]_1 = [B]_2$

$\frac{r_1}{r_2} = \left(\frac{[A]_1}{[A]_2}\right)^\alpha \rightarrow \log\left(\frac{r_1}{r_2}\right) = \alpha \log\left(\frac{A_1}{A_2}\right)$

$\frac{\log(r_1/r_2)}{\log(A_1/A_2)} = \alpha$

$r_1 = 0{,}001$
$r_2 = 0{,}002$
$A_1 = 0{,}10$
$A_2\ 0{,}20$.

$\boxed{1 = \alpha}$

Exp4
Exp5 ...

$\frac{r_4}{r_5} = \left(\frac{B_4}{B_5}\right)^\beta$

$\frac{\log(r_4/r_5)}{\log(B_4/B_5)} = \beta$

$r_4 = r_5 = 0{,}001$
$B_4 = 0{,}2$
$B_5 = 0{,}3$.

$\Rightarrow \boxed{\beta = 0}$

$r = k[A]^1[B]^0 = k[A]$.

Exp 3 $r = 0{,}003$
$[A] = 0{,}3$ $\Rightarrow \frac{r}{[A]} = k \Rightarrow k = \frac{0{,}003}{0{,}3} =$

Escaneado con CamScanner